

# AI, Post-Truth Realities, and Thai Students' Information-Seeking Behavior

*Pavel Slutskiy* | ORCID: 0000-0002-0208-9278

Faculty of Communication Arts, Center of Excellence of Communication  
Innovation for Development of Quality of Life and Sustainability,  
Chulalongkorn University, Bangkok, Thailand  
*pavel.a@chula.ac.th*

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## Abstract

This paper investigates how AI challenges traditional verification, citation, and evaluation principles, thereby eroding truth. AI-generated content, lacking verifiable citations, operates on probabilistic models that present information as probable rather than factual. The widespread reliance on AI for information-seeking risks diminishing critical evaluation, blending fact with fiction, and compromising academic integrity. To explore these effects, a mixed-method study was conducted involving 240 Chulalongkorn University students who used various online resources like Google, Wikipedia, and ChatGPT to research Rudolf Carnap's verificationism views. Their choices were analyzed to assess dependency patterns on these platforms. Further, a qualitative analysis compared AI-generated responses with verified sources to gauge their accuracy. This comprehensive approach revealed significant insights into students' preferences for information sources and the critical importance of validating AI outputs in academic settings, underscoring the nuanced impacts of digital tools on traditional knowledge standards.

## Keywords

AI-generated content – knowledge preservation – truth in the digital age – AI ethical responsibilities

## 1 Introduction

The evolution of artificial intelligence (AI) technology – in particular generative AI – is already making a massive impact across society, becoming a useful tool to automate repetitive tasks, reducing workload and potentially eliminating some specific types of work. At the forefront of this transformation are AI language models, such as ChatGPT, which have the remarkable ability to generate human-like text, answer questions, and engage in conversations. There is no doubt that these systems can do many things faster and cheaper than human beings, a fact that will have important consequences. However, even more important consequences may come from people's genuine misunderstanding of what AI can and cannot do, in particular, the ability of generative large language models (LLMs) to discern facts from falsehoods, which may challenge the long-standing concepts of education: truth and knowledge.

While AI models hold great promise for a wide range of applications, there is a growing concern about the limitations inherent in the utilization of generative AI for educational purposes. This paper analyses the relationship between AI-generated content, the concept of truth, and the preservation of knowledge, an important area for examination, as the technology may reconfigure the landscape of information ecosystems, potentially giving rise to a post-truth world where the very concept of objective and generally accepted truths becomes elusive.

Understanding the criteria for truth involves understanding the basic theories of truth and their roles in the evaluation of ideas. Theories of truth belong to the domain of epistemology, a branch of philosophy which focuses on the nature of truth and how it is determined. Epistemology concerns methods for verifying a claim and the criteria for truth assessment. The four main traditional theories of truth include: the correspondence theory, based on agreement with observable facts; the coherence theory, which relies on logical consistency within a system; the consensus theory, which looks to collective acceptance; and the pragmatic theory, which considers the practical effectiveness of a truth.

The correspondence theory suggests truth is verified through its consistency with factual reality, a principle often employed in scientific research. Conversely, the coherence theory seeks truth in the consistency and logical arrangement of beliefs or statements. The consensus theory views truth as that which is universally accepted, and the pragmatic theory measures truth by its effectiveness and applicability in real-life situations. These theories serve as critical tools for analytical thinking, assisting in the scrutiny and evaluation

of various claims and arguments, thus proving indispensable for recognizing truth across different scenarios.

Each of these theories shares roots in traditional Western epistemology wherein the concept of truth and knowledge has, for centuries, relied on the idea of verifying information by meticulously checking the premises of each statement and rigorously debating their validity. It is a concept grounded in the pursuit of facts, evidence, and a shared understanding of the world. However, as AI language models like ChatGPT enter the scene, this paradigm is being challenged. Instead of referring to factual data from reliable sources, these models predict information based on patterns and data they have absorbed, leading to an era of “probable statements” and “likely sources” rather than verified, accurate ones.

As the AI-generated content becomes more prolific and accessible, the boundaries between fact and fiction blur, giving way to what can be called the “accumulation of ambiguities.” Everything AI LLMs state appears probable, yet not entirely true. This transformation of the information landscape has important implications for the way we understand and interact with knowledge.

Our concerns extend beyond philosophical contemplation. They are rooted in the practical repercussions of a world where AI-generated content is widely utilized as a tool for the exploration of knowledge and the search for information. (Presumably, when people search for information, they search for “true” information, or at least for credible information.) What happens when individuals, academics, and institutions place their trust in AI-generated content? The consequences are profound, extending to academic research, public discourse, policymaking, and society as a whole. We are, however, particularly concerned with the use of such tools in higher education. Among multiple considerations in this paper, we will focus on the ability of AI LLMs to generate text that may appear coherent but lacks the rigorous verification and citation standards typically associated with academic research. Our concern is that students and researchers may rely on AI-generated content without scrutinising its quality, potentially leading to misinformation and compromising the credibility of academic work.

The focus on Thai university students' information-seeking behaviors is crucial due to the unique challenges and opportunities presented by Thailand's evolving educational and technological landscape. As digital literacy becomes increasingly integral to academic success, understanding how students interact with AI and online resources can inform educational policies and curriculum development. Moreover, the reliability of AI-generated content is particularly pertinent in a Thai context where educational resources and access can vary

greatly. By examining these patterns among Thai students, the study not only highlights specific cultural and educational dynamics but also contributes to a broader understanding of the global impact of AI on learning and information integrity. This insight is essential for crafting interventions that enhance critical thinking and information evaluation skills, ensuring that students are equipped to navigate and scrutinize the complex information ecosystems they encounter both academically and in everyday life.

To illustrate this concern, we offer two pieces of empirical evidence: an experiment that assesses Thai university students' information-seeking behavior and a case study that illustrates the potential pitfalls of relying on AI-generated content. We want to see if the students' information seeking patterns are changing and if they turn to AI when they need to find specific information from a particular authoritative source. The results prove our expectations to be true – students are increasingly relying on ChatGPT in their search for such information.

The second experiment aims at verifying the veracity of statements that ChatGPT provided. The results show that students trust generative AI answers as true, while the veracity verification proves the fundamental unreliability of generative AI answers. This provides us with a glimpse into the dangerous, yet already widespread among the students, illusion of truth that AI can create, effectively blurring the line between genuine knowledge and fabricated information. The consequences of such illusions may reverberate (if they are not already) not only throughout the student population but also throughout academia and the broader information ecosystem. Therefore, exploring these themes seems to be important, and we hope to contribute to the ongoing dialogue on the implications of AI-driven content generation for our understanding of truth, knowledge, and the role of technology in shaping our information ecosystem.

## 2 Literature Review

The rapid development of AI language models, particularly exemplified by models like ChatGPT, has already generated a rich body of literature and discussion around their capabilities and implications. These models, powered by deep learning technologies, have demonstrated an exceptional ability to generate coherent and contextually relevant text, making them invaluable tools in natural language processing, content generation, and even AI-driven art.



At the same time, the portrayal of AI in the media has become a subject of significant concern and scrutiny. The media often sensationalizes AI capabilities (Karanouh 2023; Leiter et al. 2023; Ouchchy et al. 2020; Pavlik 2023; Vergeer 2020), presenting them as forms of general intelligence and even sentience. The dynamics of this sensationalism highlights how the media's depiction of AI fuels public fascination and fear.

The media's narrative surrounding AI frequently takes the form of anthropomorphization, attributing human-like qualities to AI systems (Cui 2022; Deshpande et al., 2023, Scorici 2022). AI models, such as ChatGPT, are often depicted as possessing desires, intentions, and emotions, despite their operation relying solely on mathematical algorithms and statistical patterns. This anthropomorphization blurs the distinction between human cognition and AI functionality, contributing to the public's misconceptions about the true nature of AI.

The motivations driving media sensationalism of AI capabilities are multifaceted, and yet generally explainable. In digital media, where clicks and views translate directly into revenue, sensational headlines and narratives that portray AI as sentient beings prove lucrative. The clickbait culture incentivizes media outlets to generate attention-grabbing stories that attract more readers and advertisers (Molyneux and Coddington 2020). Sensationalism taps into the emotions of fear and fascination, drawing readers who are both apprehensive about a dystopian AI future and captivated by the idea of machines with consciousness. Media narratives about AI often swing between extremes of techno-optimism and techno-pessimism, catering to audiences hopeful for revolutionary advancements and those concerned about AI's unchecked power (Chan 2023). Misleading terminology compounds the issue, with media outlets employing words like "smart," "intelligent," and "conscious" when describing AI systems (Roe and Perkins 2023). These terms carry strong connotations of human-like reasoning and awareness, fostering the perception that AI operates at a level of intelligence akin to human thought. Such linguistic choices contribute to the sensationalization of AI.

Such sensationalism is driven by self-interest, greed, and the relentless pursuit of attention, but the consequences of the media's overestimation of AI's capabilities has an impact on public perception and understanding. Sensational portrayals lead people to anticipate that AI systems can perform tasks and make judgments beyond their actual capabilities, ultimately resulting in disappointment and distrust when AI falls short. Moreover, media sensationalism diminishes the nuanced understanding of AI's strengths and limitations. By blurring the line between what AI can and cannot do,

sensationalism hinders informed discussions and decision-making regarding AI integration, regulation, and ethical considerations.

In this context, factors such as accuracy and factual correctness are just one aspect of the equation. In a heavily digitally-mediated world, the paramount factor is productivity – the model's capacity to swiftly produce truth-claims on various subjects at a large scale. Critics, recognizing this capability, have already begun using terms like “poisoning,” “spamming,” and “contamination” to describe the possible effects on a future where AI-created content prevails (Heikkilä 2022; Hunger 2022).

The implications of this shift are profound, impacting not only the academic world but society at large. When AI-generated content is employed for everyday exploration of the world and the search for truth, a post-truth world begins to emerge. In this world, individuals and institutions are faced with a fundamental challenge – the very idea of objective and universally accepted truths erodes, and, in its place, probable statements and uncertainty prevail.

This transition toward a post-truth world is not solely a matter of philosophical inquiry but a practical concern with implications spanning misinformation, disinformation, and public trust. In such a world, the lines between fact and falsehood blur, rendering traditional mechanisms of knowledge validation and source citation obsolete.

Recent empirical studies have emphasized the necessity of integrating critical literacy and media literacy into educational frameworks to effectively navigate the complexities of a post-truth society. Pauncefort (2021) argues for a fundamental shift from traditional, fact-based learning to a focus on critical literacy. This approach aims to empower students with the skills needed to discern truth amidst widespread misinformation, thus prioritizing critical thinking as a key educational outcome (Pauncefort 2021).

Although we were unable to find any studies on the topic of post-truth realities and students' information seeking behaviours, previous research offers insights into related areas that are relevant to understanding the broader context. For example, a study by Xu, Feng Chen (2023) examines the differences in search performance and user experience between ChatGPT and Google Search. It highlights the trade-offs between the conversational nature of ChatGPT and the list-based results of traditional search engines like Google. Another study (Yu 2024) discusses the potential benefits and challenges of using ChatGPT in educational applications. It emphasizes the need for a bespoke application of ChatGPT, calibrated to the educational content and the disciplinary proclivities of the learners, eschewing a uniform application strategy. An interesting study by Yu, Yan, and Cai (2024) identifies the key factors that constitute the main reasons college students consistently choose

to use ChatGPT. It proposes a new research model to explain college students' satisfaction with and intention to continue using ChatGPT, asserting that ChatGPT possesses compatibility and efficiency qualities that stimulate users' perceived ease of use and usefulness.

The studies discussed highlight the importance of critically evaluating information, the need for researchers to be aware of the potential biases and limitations of their own work, and the importance of understanding the everyday information seeking behaviors of individuals. The research supports a growing academic consensus that curricula need to extend beyond traditional content delivery to include skills essential for critically evaluating information. This approach not only fosters a more informed society but also enhances democratic engagement by preparing individuals to navigate and scrutinize information sources effectively.

### 3 The Nature of AI-Generated Content

To comprehend the implications of AI-generated content on the concept of truth and knowledge, it is important to understand the mechanics and limitations of AI language models. AI language models, like ChatGPT, operate by utilizing large-scale datasets to predict text based on context and input. While they excel at generating human-like responses and are valuable for a wide range of applications, they fundamentally lack the ability to cite reliable sources as human writers do.

Historically, the pursuit of knowledge and truth has been intrinsically tied to a process of verification (see, e.g., Armstrong 1973; Pojman 2005). The concept of truth itself has been closely tied to a rigorous process of the critical evaluation of facts and sources. In academic research, and scholarly discourse in general, truth was established by the careful validation of facts and information. It relied on the credibility of the sources, the ability to cross-reference claims, and the practice of citing authoritative references. Whether through referencing credible sources, cross-referencing claims, or subjecting information to rigorous scrutiny, knowledge has been established on the foundation of factual accuracy and reliability. In this traditional paradigm, the concept of truth is considered a verifiable, testable reality (Davies 1981; McGee 1991; Russell 1912; Soames 1984; Vision 2004; Walker 1989).

AI-generated content, however, operates on a different principle. The core functionality of these models hinges on vast datasets from the internet, allowing them to predict and generate text that is contextually accurate and grammatically sound. However, the key distinction lies in their inability to

cite reliable sources and provide verifiable evidence for the information they produce in order to discern facts from falsehoods. Rather, LLMs generate text based on the patterns and data on which they have been trained, leading to outputs that resemble human language but are devoid of the traditional academic rigors of citation and verification. Instead of providing citations, they predict information based on the most likely sources, often resulting in a mix of accurate and inaccurate content. In essence, AI-generated content reflects a paradigm shift in the approach to information: predictions from data rather than references.

Such performance can be conceptualised as an “operationalization of truth” (Munn et al. 2023), where diverse, and sometimes opposing, assertions are seamlessly integrated and asserted as truths. Responses produced by LLMs emerge through a mechanical imitation of a foundational dataset in a probabilistic manner, incorporating sufficient variability to imply autonomy, yet entirely governed by the mimicry of linguistic patterns present within that dataset. Bender et al. (2021) thus argue that LLMs can be described as “stochastic parrots”: They are good at replicating human speech and intellect, yet possess no genuine comprehension of the meanings behind these words and concepts. The rapid generation of claims and their inundation into information spaces creates a unique form of epistemic hegemony, magnified by artificial intelligence. This operationalization of truth underscores that veracity is not an inherent and timeless concept but an actively constructed one.

A side effect of mimicking these probabilistic sequences is that LLMs often propagate prevalent errors, as highlighted by Ouyang and colleagues in their 2022 study on training language models through human feedback (Ouyang et al. 2022). The tendency for these models to echo a statement increases with its repetition in the data, leading to “common token bias.” Consequently, LLMs perpetuate errors in reasoning that have been recognized by traditional philosophy, essentially echoing the most frequently stated views and neglecting the biases inherent in their perspectives. This “common token bias” underscores the limitations of using consensus as a measure of truth (see Yu et al. 2023).

While these models excel at recognizing statistical patterns in large datasets, they fundamentally lack genuine consciousness and contextual understanding of the concept of truth. AI models generate responses by identifying statistical associations between words and phrases in their training data. They assess the probability that a particular word or phrase follows or co-occurs with another. This probabilistic approach drives their ability to generate coherent text, but AI models do not possess a conscious understanding of it. They do

not engage in the dynamic adaptation that characterizes real comprehension. Instead, they rely on pre-established patterns and associations to produce text that appears contextually relevant. This is why it can produce outcomes that may superficially resemble correctness but are ultimately inaccurate. This same principle explains why Language Models like ChatGPT often provide responses to historical inquiries that seem plausible on the surface but are factually incorrect. LLMs lack a true comprehension of what constitutes a fact. While they do contain terms such as “fact” in their distributional semantic data, at their core, they consist of interconnected words, each defined by its close associations with other words. In essence, LLMs do not possess a genuine understanding. They are engineered tools that generate probabilistic responses based on the relationships between terms in their training data. ChatGPT, for instance, does not possess the capacity to discern when its response does not fulfill a given prompt, as it lacks a monitoring mechanism. It lacks the kind of rational cognition required to recognize that it has given an incorrect response to the prompt. If ChatGPT were capable of “knowing” when it had answered the prompt incorrectly, it would not have produced an incorrect response in the first place (see deBoer 2023).

These challenges inevitably lead to concerns about reliability. While AI models are trained on vast datasets, they are not infallible and can generate erroneous or misleading information (also known as “hallucinations”, see Salvagno et al. 2023; Alkaissi and McFarlane 2023). In fact, one can argue that for the purposes of creative expression, marketing endeavors, and sensational journalism, the allure of language models lies exactly in their capacity to take “creative liberties” with truths. In these contexts, the bending of truth is inherent if not desirable, as societal or genre norms recognize the entertainment value, distraction, or even comfort that such non-true content can provide. Truth becomes a malleable entity, shaped to fit the narrative's demands. Yet, this flexibility encounters severe limitations in critical sectors like academia, healthcare, transport safety, and the judicial system, where the tolerance for inaccuracy and falsehood is significantly lower.

The concept of “tolerance” in this context is a meta-truth, delineating the permissible boundaries within which a language model operates. Some situations demand a nuanced, probabilistic truth, shaded in gray, while others insist on stark black-and-white clarity. Until these situations have been effectively designed, the potential consequences of widespread reliance on AI for information-seeking purposes are potentially very significant. In a world where AI-generated content is readily accessible and often indistinguishable from human-generated content, individuals and institutions may be tempted to embrace the convenience and efficiency offered by AI models.

The convenience of AI-generated content may lead users to accept information at face value without critically evaluating its sources or accuracy. That is, users might forego the traditional due diligence of verifying information, trusting AI-generated content implicitly. The blurring of lines between AI-generated content and human-authored content can lead to a scenario where everything is rendered as probable, yet not entirely true, and where information is accepted based on presentation rather than verifiability. In this context, the distinction between fact and fiction becomes increasingly challenging.

As AI-generated content becomes a tool for the exploration of the world the search for true reliable information, it opens up a door to a post-truth world, the notion of which has emerged as a compelling and, in many ways, disconcerting framework to understand the contemporary education landscape.

The consequences of widespread reliance on AI for education purposes are significant, impacting critical thinking, the verifiability of information, and the integrity of knowledge in academic and intellectual domains. These consequences extend to academic research, as well as other intellectual pursuits. In these contexts, the reliance on AI-generated content without proper validation can erode the foundations of knowledge, impacting the quality and integrity of scholarly (and not only scholarly) work. Scholarship and research increasingly grapple with challenges related to the verifiability and credibility of information, which means that academic institutions may need to adapt to address these challenges.

#### 4 Experiment: AI's Impact on Information-Seeking Behavior of Thai University Students

To support our assertions, we conducted a study which explores the information retrieval strategies employed by undergraduate students. Information retrieval is a fundamental component of academic research and plays a pivotal role in shaping students' understanding of complex topics. To guide the exploration of Thai university students' information-seeking behavior in the context of AI-generated content and traditional search engines, this study seeks to answer the following research questions:

- 1) What are the predominant information sources utilized by Thai university students when confronted with tasks requiring the retrieval of unfamiliar academic content? This question aims to identify whether students prefer traditional search engines like Google, database-driven resources like Wikipedia, or AI-driven tools like ChatGPT.

- 2) How accurate is the information retrieved from AI-generated content? This question investigates the reliability of AI-generated responses and evaluates them against the established benchmarks of accuracy typically found in more traditional academic resources.

A total of 240 undergraduate students from Chulalongkorn University (Bangkok) were recruited for this study. All participants were given access to the internet and were assigned the task of quickly discovering what Rudolf Carnap thought about verificationism and supporting a summary of his ideas with a quote from one of his works. This topic was chosen for its relevance in the context of philosophy and epistemology, and the availability of various online resources.

Participants were told that they could use any online resources, including search engines, general-purpose AI chatbots, and encyclopedic sources. Immediately after completing the research task, they were required to report where they found the relevant information via a structured digital questionnaire designed to collect precise data on the students' information sources, such as Google, Wikipedia, or ChatGPT (including the exact version). This method ensured a consistent and systematic approach to gathering data across all participants, providing a reliable basis for analyzing their behaviors. The use of a structured questionnaire also helped mitigate the risks of recall bias, which can occur in less controlled data collection methods such as retrospective interviews or informal chat logs.

The findings of this study demonstrate varying information retrieval strategies among undergraduate students. Of the 240 participants, 57% reported using Google as their primary source of information, reflecting the conventional use of search engines for academic research. In contrast, 22% of the participants sought information from ChatGPT (version 3.5), indicating a preference for AI-powered conversational agents. The remaining participants (21%) relied on Wikipedia as their primary source (Figure 1).

The choice of ChatGPT by a significant proportion of students suggests an inclination towards artificial intelligence as a source of information. Participants who used ChatGPT reported its convenience in providing direct and concise answers to their queries.

Although Google remains a popular choice for its convenience and familiarity, ChatGPT's inclusion indicates the growing integration of AI-based platforms in academic research. The results confirm that the connection between the emergence of AI chatbots like ChatGPT as a source of information and the evolving landscape of information-seeking behavior.

The second part of the experiment addressed our previously discussed concerns about the need to verify the accuracy of information provided by

■ Google      ■ ChatGPT      ■ Wikipedia

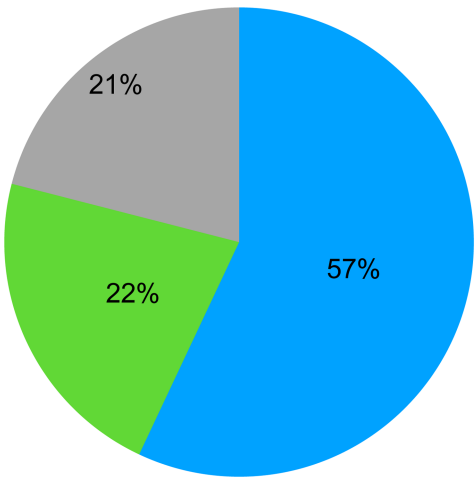


FIGURE 1 Students' information retrieval strategies

AI chatbots in order to illustrate the real-world implications of relying on AI-generated information and the challenges it presents in the pursuit of knowledge.

To illustrate the practical implications of AI-generated content in the quest for knowledge, we collected the quotes from the ChatGPT responses to the first query – to describe Rudolf Carnap’s thought about verificationism and to support the summary of his ideas with a quote from one of his works. In total, 37 of student responses included a direct quote attributed to Carnap.

The AI responses collected from the student submissions appeared to provide an informative and credible answer, mentioning that Carnap discussed verificationism and the principle of empirical significance in his works. The answers also stated that later Carnap modified some of his views to accommodate criticisms and concerns about the strictness of verification.

The responses which included a quote attributed to Carnap encapsulated his perspective on verification and meaning. Some of these responses read as follows:

The goal of the new logic is the separation of purely formal relations in which the meaning of the sentences is not involved, from those which concern meaning, and therefore belong to the realm of empirical knowledge.

“The Logical Structure of the World”, 1928



The meaning of a statement is the method of its verification.

"The Logical Structure of the World", 1928

The verification of the proposition that an entity of a certain kind exists consists in the successful application of the methods of empirical science which lead to the finding of an entity of this kind.

"The Logical Structure of the World", 1928

The task of philosophy is to bring about a clearness and logical order in the knowledge which we possess, to make the structure of this knowledge transparent, to analyse the meaning of the concepts which occur in it.

"The Elimination of Metaphysics Through Logical Analysis of Language", 1932

In science there are no 'depths'; there is surface everywhere: all experience forms a complex network, which cannot always be surveyed and, therefore, can be grasped only step by step.

"Logical Foundations of Probability", 1950

In the field of the empirical sciences, more particularly in the field of physics, we have to do with a system of statements which are so interconnected that every statement can be deduced from the others. This system is not a deductive system in the strict sense of the term, for its statements are not all analytic but synthetic.

"Philosophy and Logical Syntax", 1935

The task of the logic of science is to analyze the conceptual and theoretical structure of science.

"The Logical Syntax of Language", 1934

Every statement about reality aims to express a certain structure; and we use the linguistic expression of this structure as the criterion for the truth or falsehood of the statement.

"The Logical Syntax of Language", 1934

Upon verifying these quotes, however, we discovered a striking (although somewhat predictable) pattern: the quotes attributed to Carnap were entirely absent from the cited works. They were, in actuality, AI-generated constructs – mere imitations of Carnap's writing style and ideas.

These results are both enlightening and disconcerting. The AI-generated responses bore the guise of a legitimate scholarly reference, presenting

themselves as a reliable source of information. In an academic context, the inclusion of a quotation from a prominent philosopher carries significant weight and would ordinarily be incorporated into research and scholarly discourse.

Following the experiment, we conducted a debriefing session, which included a discussion of the reliability of the answers that the participants submitted. All 37 of the students who included ChatGPT quotes were confident that they were illustrating the summary of Carnap's ideas with a direct citation from one of his works.

This case serves as an example of the illusion of truth that AI-generated content can create and its potential impact on education. The transformation of AI-generated content from a tool for creative expression to a tool for knowledge-seeking necessitates a careful reevaluation of how we navigate this evolving landscape. In a world increasingly dependent on AI-generated content for research and knowledge acquisition, the consequences are profound. Had the fabricated quote gone unverified, it could have perpetuated a fallacious ethical stance. Students (as well as scholars and researchers) relying on such an erroneous reference might have unwittingly disseminated inaccurate information, affecting the course of discussions. Over time, this misrepresentation could embed itself deeply within the academic discourse, making it increasingly challenging to rectify and undermining the integrity of scholarly endeavors and ethical considerations within the field. This simple experiment underscores the pressing challenge posed by AI-generated content and highlights the necessity for a diligent approach in knowledge acquisition.

## 5 Discussion and Recommendations

The emergence of AI language models like ChatGPT raises a fundamental ethical question: Who bears responsibility for the content these models generate and for how this content is utilized? Is it the developers who design and train the models, the users who rely on them, or both? As AI-generated content becomes increasingly integrated into academic, professional, and personal spheres, addressing these questions becomes increasingly important.

At the core of the ethical discourse lies the duty to ensure that the technology is used responsibly and ethically. The responsibility extends to the transparency of AI systems, the quality of the data they are trained on, and the accuracy of the outputs they produce. Of course, developers play a crucial role in minimizing the potential for misinformation and actively preventing the generation of deceptive or fabricated content. But users of AI-generated content, especially in academic or research settings, have a shared responsibility to approach

AI-generated information with discernment and skepticism. Users should be vigilant in critically evaluating the information provided and cross-referencing it with established knowledge and reputable sources.

To address the challenges presented by AI-generated content in academic research without relying on government regulation, a multifaceted approach is recommended. Although we do not have a universal solution to these complex challenges, some recommendations can be proposed. Central to this solution is the principle of informed decision-making, which empowers users to critically evaluate and discern the nature of the information they encounter.

One solution could be the implementation of a comprehensive disclaimer system on AI-generated content platforms. AI developers can enhance transparency by implementing features that clearly indicate the limitations of the content generated by AI. Encouraging AI platforms to provide information about the source of information (or inability to provide accurate citations) can improve transparency. This disclaimer would emphasize the inherent limitations of AI systems in verifying information, cross-referencing sources, or providing citations. It serves as a crucial reminder that AI-generated content should not be relied upon as authoritative or factual, especially in academic and research contexts, where accuracy and reliability are paramount.

In addition, the system would provide suggested use recommendations, offering guidance to users on how best to leverage AI-generated content. While it may encourage creative inspiration or brainstorming, it would simultaneously discourage heavy reliance on AI-generated content for critical academic research or fact-checking. These recommendations provide users with a framework for responsible and effective utilization of AI-generated information.

To increase the effectiveness of such a warning system, close collaboration with educational and academic institutions is vital. These institutions would play a pivotal role in educating students about the potential challenges associated with AI-generated content and underscoring the significance of cross-referencing and critical evaluation, equipping individuals with the tools needed to navigate this evolving landscape adeptly. Generally, educational institutions can play a pivotal role by integrating digital literacy and critical thinking skills into their curricula. Workshops and seminars on AI literacy should be organized to increase students' awareness and knowledge about AI-generated content's potential implications. These skills can empower students and researchers to effectively evaluate information, including AI-generated content.

In addition to this, organizations, including universities and academic publishers, should establish clear guidelines for the ethical use of AI-generated content that emphasize the importance of verification and cross-referencing AI-generated information with established sources. Ethical codes can be

integrated into academic publishing and research practices. Peer review and cross-referencing should also be promoted as a means of maintaining academic rigor. Scholars and researchers should be encouraged to critically evaluate the information, including any quotes or citations, and ensure that they align with established knowledge and reputable sources.

Developing specialized AI verification tools can further assist researchers in assessing the accuracy and reliability of AI-generated content. Interdisciplinary collaboration between experts in AI, ethics, and the relevant academic domain is essential for addressing ethical and epistemological concerns related to AI-generated content in specific fields. Reinforcing academic integrity policies to include guidelines for using AI-generated content can contribute to a more ethical academic environment.

Lastly, fostering open dialogue between AI developers, educators, researchers, and students is essential to address emerging challenges and develop best practices for AI use in academic contexts. This comprehensive approach can help mitigate the ethical concerns associated with AI-generated content in academic research.

Overall, the continued development and refinement of AI algorithms in the future should enhance transparency and accuracy while reducing the generation of deceptive or misleading content. We have little doubt that the evolution of AI technology will see substantial improvements, with the potential to minimize errors that currently occur. These processes will elevate the quality of AI-generated information and ensure that users can trust its reliability.

While it is reasonable to acknowledge that generative AI may exhibit advancements in accuracy, it is essential to recognize a fundamental reality. Achieving correctness in outputs where previous errors occurred does not alter the intrinsic nature of AI systems. This distinction often remains elusive to some, possibly stemming from a reluctance to acknowledge it. The central premise is that the precision and reliability of a machine learning system's output do not serve as evidence of its capacity for thought, or consciousness, or understanding of what a fact is. These are distinct and unrelated considerations.

When confronted with observations that contemporary "AI" systems do not emulate human thought processes, proponents of AI advancements emphasize the improved results and accuracy achieved, as if these achievements address the underlying criticisms. However, it is crucial to understand that they do not address the core issues at hand. We do not deny nor negate the impressive capabilities of LLMs and similar probabilistic AI-like models. They are undeniably capable of performing remarkable tasks. Nevertheless, it is important not to ascribe to them capabilities beyond their intended functionalities.

## 6 Conclusion

Today's education system is at a critical phase of adaptation to the new technology, where human knowledge converges with artificial intelligence. The appeal of AI-generated content lies in its capacity to emulate true statements, yet it notably lags in verification and accuracy. This disparity holds the potential to undermine the bedrock of knowledge, blurring the distinction between reality and fiction, thereby propelling us into a post-truth era where truth itself becomes elusive.

This however should not be a cause for another moral panic. Although the challenges may seem formidable, a reasonable response seems rather obvious. It is important to be realistic about the limitations of generative AI and not to overhype its potential which could lead to unrealistic expectations and over reliance on it. Of course, there is also a matter of common sense responsibility shared among developers, users, and broader stakeholders. Developers should place a premium on transparency, data quality, and precision in AI models. Users must exercise discernment and apply critical thinking when engaging with AI-generated content, particularly in academic and research domains. These skills probably need to be taught as a part of new media literacy skillset. The proposed disclaimer system, including usage recommendations, would also empower users to make informed choices about the content they encounter. This seems to be a practical solution that advances the cause of knowledge preservation, accountability, and transparency in the 6-infused landscape.

Overall, the cultivation of literacy and the proactive implementation of a warning system on AI-generated content platforms that was proposed in this paper seem to offer at least a temporary solution to the problem safeguarding the core principles of truth and knowledge integrity in education. The pursuit of knowledge endures as a quintessential human endeavor, and even amid technological strides, our duty is to protect the truth.

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